

Spatio-temporal heterogeneity of gas concentrations and its propagation into ventilation and emission estimates in a naturally ventilated dairy barn

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Abstract

Gas concentrations in naturally ventilated dairy barns are spatially heterogeneous, but the extent to which sampling height, coverage and density propagate into calculated ventilation and emissions is rarely quantified. Carbon dioxide (CO₂), methane (CH₄) and ammonia (NH₃) were measured sequentially at 51 internal sampling points arranged as 17 vertical triplets in Campaign 1. Following technical failure of the two Fourier-transform infrared analysers, Campaign 2 used a pragmatic 32-point top-and-bottom network connected to two cavity ring-down spectrometers. Instrument scales in Campaign 1 were harmonised using a 24-h co-location test. Observed sampling-point distributions, mixed-effects models, gas ratios and Shannon entropy were used to characterise heterogeneity. A standard-input indirect CO₂-balance calculation then propagated alternative spatial networks into ventilation rate and CH₄ and NH₃ emissions. Gas, height, season and aggregation scale all affected apparent precision. Top concentrations exceeded bottom concentrations by 1.7, 3.1 and 4.5% for CO₂, CH₄ and NH₃, respectively, in Campaign 1, and by 3.3, 7.4 and 22.8% in Campaign 2. The middle height was the closest single-height surrogate for the other two heights for all concentrations and ratios. SP17 was statistically most representative overall, whereas the pragmatic central SP25 ranked fourth and was not universally representative. Against the 51-point balance, the 34-point top-and-bottom network produced median absolute errors of 1.2% for ventilation, 1.4% for CH₄ emission and 2.1% for NH₃ emission; the Campaign-2-like 32-point network produced 1.7, 1.6 and 2.8%, respectively. Thus, reduced spatial coverage can preserve campaign-scale estimates when both vertical extremes remain

represented, but a single height or point cannot be assumed representative across
gases.

Keywords: ammonia; dairy housing; methane; mixing ratio; sampling height; spatial
coefficient of variation

Nomenclature

Symbol	Definition
CRDS	cavity ring-down spectroscopy
CV	coefficient of variation, %
FTIR	Fourier-transform infrared spectroscopy
NVDB	naturally ventilated dairy barn
RE	relative error against the contemporaneous dense-network mean, %
SD	standard deviation
SP	sampling point

1 Introduction

Emission estimates for naturally ventilated livestock buildings depend jointly on airflow and the concentration field sampled across that airflow. Wind, buoyancy, animals, manure surfaces and internal obstructions produce spatially variable transport, so analytical accuracy alone does not ensure spatial representativeness (Calvet et al., 2013; Ogink et al., 2013). Direct wind-tunnel measurements demonstrated that combining concentration and velocity at one constant height could produce emission errors up to 97%, whereas vertically resolved sampling reduced the error to below 5% in the tested cases (Janke et al., 2020b). A concentration sensor should therefore be evaluated at the spatial and temporal resolution at which it will be used.

Vertical gradients are gas- and site-specific. Sahu et al. (2023) measured significant height effects for CO₂, CH₄, and NH₃ at the outlet of a naturally ventilated dairy building and commonly observed greater concentrations near the top. In another open dairy barn, NH₃ was greatest near the floor whereas CH₄ was greatest at the upper level, demonstrating that a universal vertical ordering cannot be assumed (D’Urso et al., 2024). Horizontal sampling density also affects inferred means and variability (D’Urso et al., 2022; De Voeleer et al., 2017). Dombia et al. (2021) further showed that the frequently used approximately 3 m sampling height represented well-mixed conditions only for a subset of simulated convection cases. These results motivate measurements both near the customary height and at uncommon vertical extremes.

Temporal aggregation is equally important. Rapid changes in wind and thermal forcing may create large short-term spatial differences that are attenuated in daily or campaign means. Wind direction can also change which openings are inlets or outlets and which indoor locations are representative (Saha et al., 2013; König et al., 2018; Janke et al., 2020a). Regional meteorological observations describe the synoptic forcing but do not replace velocity measurements at the barn openings. Consequently, seasonal comparisons should quantify temporal evolution while avoiding causal claims that cannot be resolved without co-located airflow measurements.

Gas ratios provide a complementary diagnostic. If all gases respond identically to dilution, their ratios should be spatially more stable than their concentrations. Persistent gradients in NH_3/CO_2 , CH_4/CO_2 , or NH_3/CH_4 instead indicate differences in sources, sinks or transport. Chen et al. (2026) used entropy to describe the sensitivity of CFD airflow fields to prescribed wind cases. Here, entropy instead describes temporal state occupancy in sequential field concentrations and is interpreted as a complementary, gas-specific measure of location informativeness.

Two hypotheses were tested: (H1) the precision of each gas differs across spatial and temporal resolutions; and (H2) gas concentrations generally increase with increasing vertical sampling height, as suggested by previous measurements (Sahu et al., 2023). Secondary objectives were to identify representative heights and individual SPs, quantify gas coupling and entropy, and determine how sampling height, coverage and density propagate into indirect CO_2 -balance ventilation and emission estimates. SP25 was evaluated as a pragmatic central candidate, not specified as the statistical optimum.

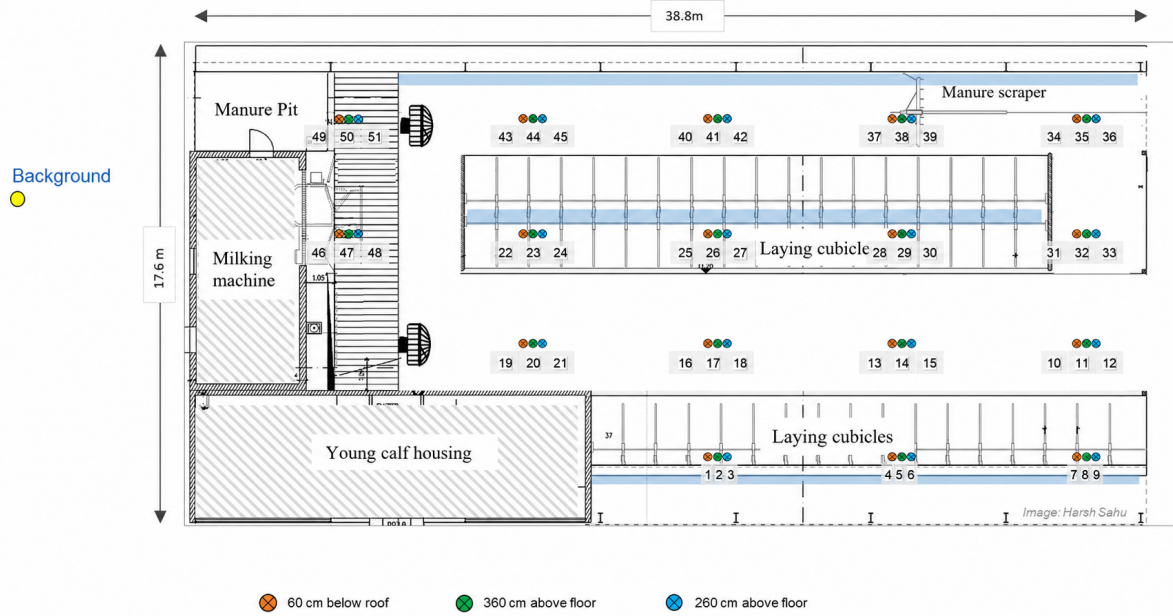
2 Materials and methods

2.1 Barn and sampling layout

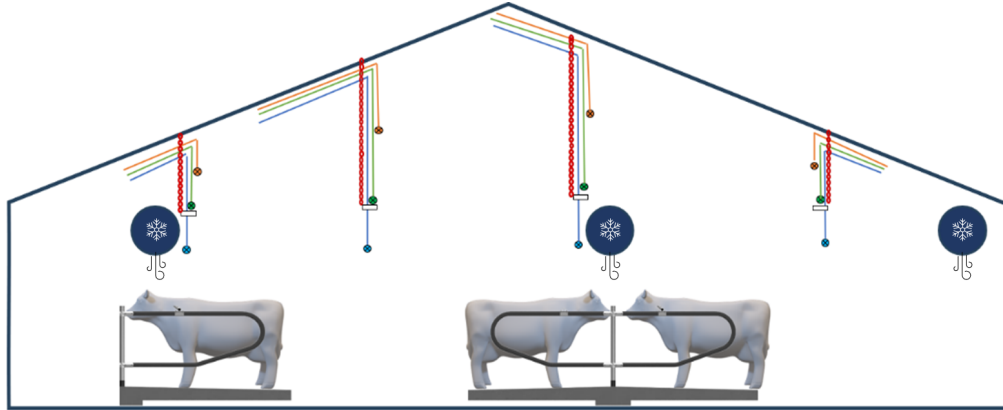
Measurements were conducted in an occupied NVDB at Gross Kreutz, Brandenburg, Germany. The barn was 38.88 m long and 17.65 m wide, with an internal airspace of approximately 4529 m^3 and capacity for 58 dairy cows. Seventeen horizontal positions covered lying, feeding, walking and manure-adjacent zones (Fig. 1). At every position, a vertical triplet comprised a top SP 0.60 m below the local roof, a middle SP 3.60 m above the floor, and a bottom SP 2.60 m above the floor. Thus, the middle and bottom layers were separated by only 1.00 m; the top layer followed the sloping roof and represented an intentionally uncommon near-roof position.

2.2 Campaigns and gas measurements

Campaign 1 combined dense observations from June–August 2024 (summer; 90 measurement dates) and 1–24 October 2024 (autumn; 24 dates). All 51 internal SPs were



(a) Plan view and 51 internal SPs.



(b) Vertical arrangement of the sampling tubes.

Figure 1: Experimental barn and sampling design. Location numbers increase in vertical triplets: top, middle and bottom. The plan also shows functional zones and mechanical mixing fans.

measured with two CRDS and two FTIR analysers. During the experiment the FTIR instruments developed technical defects. Campaign 2 was therefore not a data-driven deletion experiment: the network was pragmatically reconfigured to fit the 32 available CRDS ports. The middle layer was removed, while top and bottom points were retained except SP19 and SP40. Campaign 2 was measured from 16 November to 31 December 2024 (15 late-autumn and 31 winter dates). Sparse FTIR-connected observations at SP49 and SP51 were retained in the data audit but excluded from Campaign 2 inference, leaving 32 eligible CRDS points. Campaign 1 is consequently the principal dense experiment and Campaign 2 an extended seasonal test of persistent top–bottom variation. Sampling proceeded sequentially at four-minute resolution. The first 60 s after switching was discarded and the remaining 180 s were averaged; the time stamp represented the end of the averaging interval.

A preliminary 24-h co-located field comparison was performed before Campaign 1. Two polytetrafluoroethylene sampling lines terminated at the same central barn position; one line was divided between the CRDS instruments and the other between the FTIR instruments. FTIR values were 6% higher for CO₂ and CH₄, and 9% higher for NH₃. Campaign 1 FTIR values were therefore harmonised as $C_{\text{CO}_2,\text{corr}} = C_{\text{CO}_2,\text{raw}}/1.06$, $C_{\text{CH}_4,\text{corr}} = C_{\text{CH}_4,\text{raw}}/1.06$, and $C_{\text{NH}_3,\text{corr}} = C_{\text{NH}_3,\text{raw}}/1.09$. CRDS values were unchanged. Only missing, zero or negative gas values were excluded; no lower positive-concentration threshold was imposed.

2.3 Meteorological context

Hourly temperature, relative humidity, precipitation, wind speed and wind direction were obtained from the German Meteorological Service for the nearest suitable station and aligned in UTC (Deutscher Wetterdienst Climate Data Center, 2024). Wind roses and temperature–humidity time series were used to describe seasonal forcing. Westerly flow was defined as 225–315°. Hourly internal concentration means under westerly and other directions were compared using Wilcoxon tests, and exploratory mixed models included wind speed, direction class, temperature, relative humidity and precipitation. These analyses describe association, not barn-scale causation, because velocity and direction were not measured simultaneously at every opening.

2.4 Responses and statistical analysis

Ratios were calculated on paired observations as

$$R_{a/b} = C_a/C_b. \quad (1)$$

Spatial precision within each hour, day, ISO week and campaign period was expressed as

$$CV_{g,t} = 100 \frac{SD_i(C_{g,i,t})}{\overline{C}_{g,t}}, \quad (2)$$

where i is SP and t is aggregation interval. Because ratio distributions can be sensitive to small denominators, medians and interquartile ranges of block-level CVs were emphasised alongside campaign means and SDs. A mixed model tested log-transformed CV with response, temporal scale and their interaction as fixed effects and period as a random intercept.

For dense Campaign 1 observations, log response was modelled with height, period and their interaction as fixed effects and horizontal position and date as crossed random intercepts. A second model used only locations shared by both campaigns and tested top versus bottom across four periods, again with horizontal position and date as random effects. Tukey-adjusted estimated-marginal-mean contrasts were calculated. The consistency of H2 was further expressed as the number of horizontal positions at which $\overline{C}_{\text{top}} > \overline{C}_{\text{middle}} > \overline{C}_{\text{bottom}}$.

For complete two-hour dense-network blocks, each single-height mean was compared with the 51-location mean:

$$RE_{h,b} = 100 \frac{\overline{C}_{h,b} - \overline{C}_{\text{dense},b}}{\overline{C}_{\text{dense},b}}. \quad (3)$$

Shannon entropy was calculated after discretising each response into four equiprobable bins within period. The resulting tiles assess empirical compositional information at a location; they are not estimates of airflow or local mean age of air. Analyses were conducted in R 4.5.3. Significance was assessed at $\alpha = 0.05$, with effect sizes and temporal consistency interpreted alongside p -values.

Representative heights were evaluated in complete Campaign 1 two-hour blocks by comparing the median at each height with the median of the other two heights. Candidate SPs were ranked by leave-one-SP-out agreement with the contemporaneous median of the remaining network, combining normalised median absolute error and Spearman correlation across the three concentrations and three ratios. SP25 was retained as a pragmatic central comparator, while the ranking was allowed to identify a different statistical optimum.

For an emission-sensitivity analysis, the standard-input indirect CO₂-balance function was applied to matched Campaign 1 blocks. The 51-SP network was the reference. Alternative estimates used each 17-point height, the 34-point top-and-bottom network, and a 32-point network matching the Campaign 2 coverage. Balanced density experiments retained all three heights at randomly selected horizontal positions (3, 6, 12, 24, 36 or 51 SPs; 100 fixed networks per density). Only missing, non-positive or otherwise undefined

results were excluded from their respective calculations. Because an inverse ΔCO_2 balance becomes unstable as the indoor–outdoor difference approaches zero, estimates with $\Delta\text{CO}_2 \geq 200$ ppm formed the primary analysis; 70–199 ppm was retained as a sensitivity band and values below 70 ppm were flagged as unreliable rather than deleted from the processed data. The 200 ppm rule is deliberately conservative and reflects published limitations of short-period CO_2 -balance calculations (Pedersen et al., 2008). The calculated rates are standard-input equivalents used to compare spatial configurations, not definitive whole-barn flux measurements, because matching opening velocities were unavailable.

3 Results

3.1 Gas-specific precision across temporal resolutions

The analysis included 159,866 internal records: 108,078 in summer Campaign 1, 31,156 in autumn Campaign 1, 8,906 in late-autumn Campaign 2 and 11,726 in winter Campaign 2. Precision differed by response, temporal scale and their interaction (all $p < 0.001$). At hourly resolution, median spatial CV was 7.5–8.9% for CO_2 , 22.0–27.2% for CH_4 , and 24.3–33.2% for NH_3 . Daily medians decreased to 4.4–6.1%, 9.5–17.1%, and 19.9–29.3%, respectively. Weekly values were similar to or below daily values for CO_2 and CH_4 , whereas NH_3 remained comparatively heterogeneous. H1 was therefore supported: both gas identity and aggregation scale altered apparent precision.

Mean concentrations also changed over time. Across heights, CO_2 and CH_4 were generally higher in autumn and winter than in summer, while NH_3 declined from approximately 2.1 ppm in summer to below 1 ppm in Campaign 2. The ratios did not track concentration changes identically, confirming that temporal variation was not a uniform multiplication of all gases.

3.2 Vertical structure and its seasonal dependence

In dense Campaign 1 data, height, period and their interaction affected every concentration and ratio (all interaction $p < 0.001$). Summer gradients were small: geometric top means exceeded bottom means by approximately 0.9% for CO_2 , 1.1% for CH_4 , and 1.6% for NH_3 . In autumn, corresponding contrasts increased to approximately 4.7%, 10.8%, and 15.9%. The strict top–middle–bottom ordering occurred at only 9/17, 10/17, and 10/17 positions for the three gases in summer, but at 13/17, 13/17, and 14/17 in autumn. H2 was therefore conditionally, not universally, supported.

The four-period top–bottom models confirmed significant height-by-period interactions for all responses ($p < 0.001$). Top concentrations exceeded bottom concentrations significantly in every period except CO_2 in late autumn. The largest proportional separation occurred for NH_3 , especially in autumn and winter. Ratio gradients were also

Observed gas concentrations and ratios at each sampling point

Campaign 2 contains the 32 statistically eligible CRDS points; FTIR-connected SP49 and SP51 are excluded

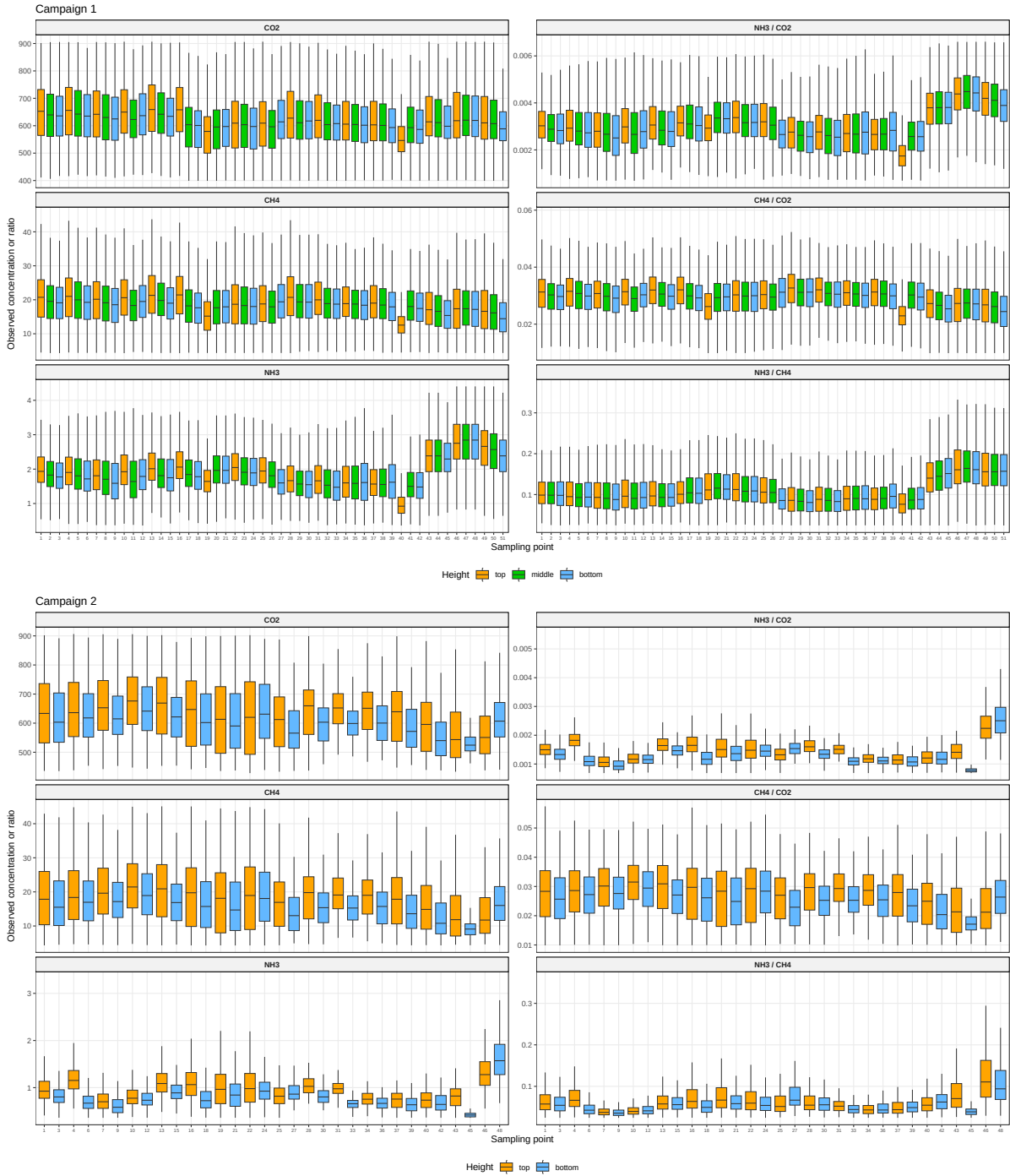


Figure 2: Observed concentrations and dimensionless ratios at every statistically eligible SP. Campaign 1 contains all 51 internal SPs; Campaign 2 contains the 32 CRDS-connected top and bottom SPs. Boxes show medians and interquartile ranges, with whiskers extending to 1.5 times the interquartile range. Colours denote top (orange), middle (green), and bottom (blue). The figure displays observations rather than model-derived estimates.

significant, showing that common ventilation-driven dilution alone could not explain the vertical ordering.

The middle-only distribution lay closest to the other-height median for every response. Median absolute differences were 1.30, 2.97 and 3.11% for CO₂, CH₄ and NH₃, compared with 2.05–5.36% for the bottom and 2.25–5.88% for the top height. Thus, the middle height was statistically most representative. Its omission from Campaign 2 resulted from FTIR failure and the 32-port CRDS constraint, not proof that it was unnecessary.

3.3 Representative height, sampling point and gas coupling

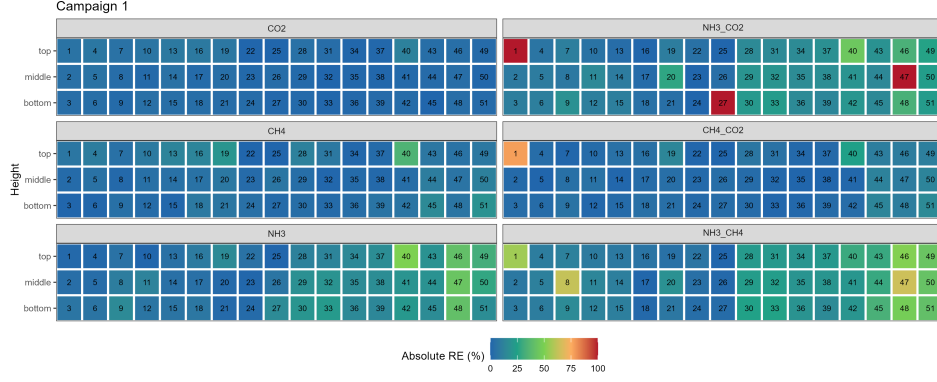
The leave-one-SP-out ranking identified SP17 (middle height) as the best overall single-point surrogate, followed by SP26 and SP14, also at the middle height. SP25 ranked fourth. Its central location remains operationally attractive, but the result rejects a universal claim that geometric centrality guarantees statistical representativeness. Middle-height points dominated the leading ranks, consistent with the height-level comparison (Fig. 4).

Observed CH₄–CO₂ coupling was strong at all heights (Campaign 1 $R^2 = 0.72$ – 0.82 ; Campaign 2 $R^2 = 0.94$), whereas NH₃–CO₂ coupling was weak in Campaign 1 ($R^2 = 0.07$ – 0.10) and only moderate in Campaign 2. The linear patterns therefore support common dilution for CH₄ and CO₂ more strongly than for NH₃, but do not prove complete physical mixing (Fig. 5).

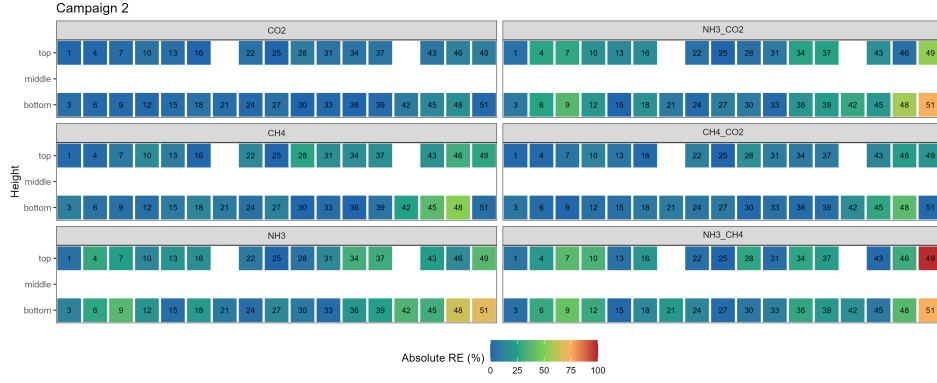
3.4 Entropy and meteorological context

Entropy varied by gas and position (Fig. 6). Locations with high entropy for one gas were not invariably high for another, consistent with gas-specific source and transport behaviour. The result did not identify a universal sensor position and should not be equated with CFD-based optimisation.

Seasonal wind roses and temperature–humidity records showed that the campaigns sampled different external regimes (Fig. 7). Westerly hours were associated with lower, rather than higher, internal hourly mean concentrations in the unadjusted summaries: for example, autumn Campaign 1 means changed from 671 to 601 ppm CO₂, 21.3 to 17.2 ppm CH₄, and 1.76 to 1.37 ppm NH₃. Winter differences were also negative. These associations are consistent with enhanced ventilation under some westerly regimes, but regional wind cannot distinguish dilution from plume transport or opening-specific flow.



(a) Campaign 1.



(b) Campaign 2.

Figure 3: Absolute relative error of each location mean against the pragmatic central reference, SP 25. Ratios were calculated without percentage scaling; only their relative errors are percentages. All panels use the same 0–100% colour scale.

Campaign 1 representative height and sampling point

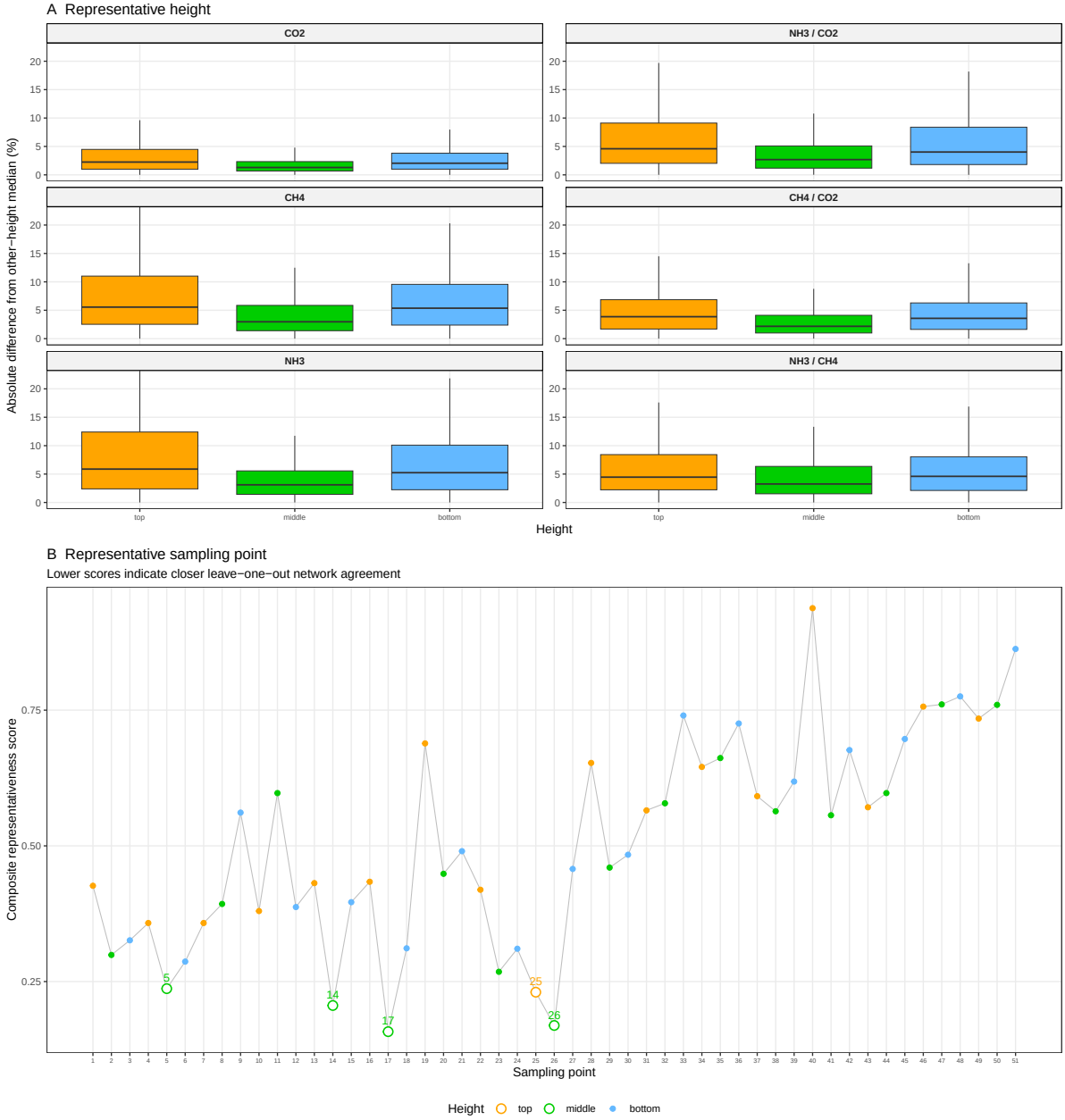


Figure 4: Campaign 1 representativeness. (a) Absolute difference between each height median and the median of the other two heights in matched two-hour blocks. (b) Composite leave-one-SP-out score across three concentrations and three ratios; lower values indicate closer agreement with the remaining network. The five best SPs are labelled.

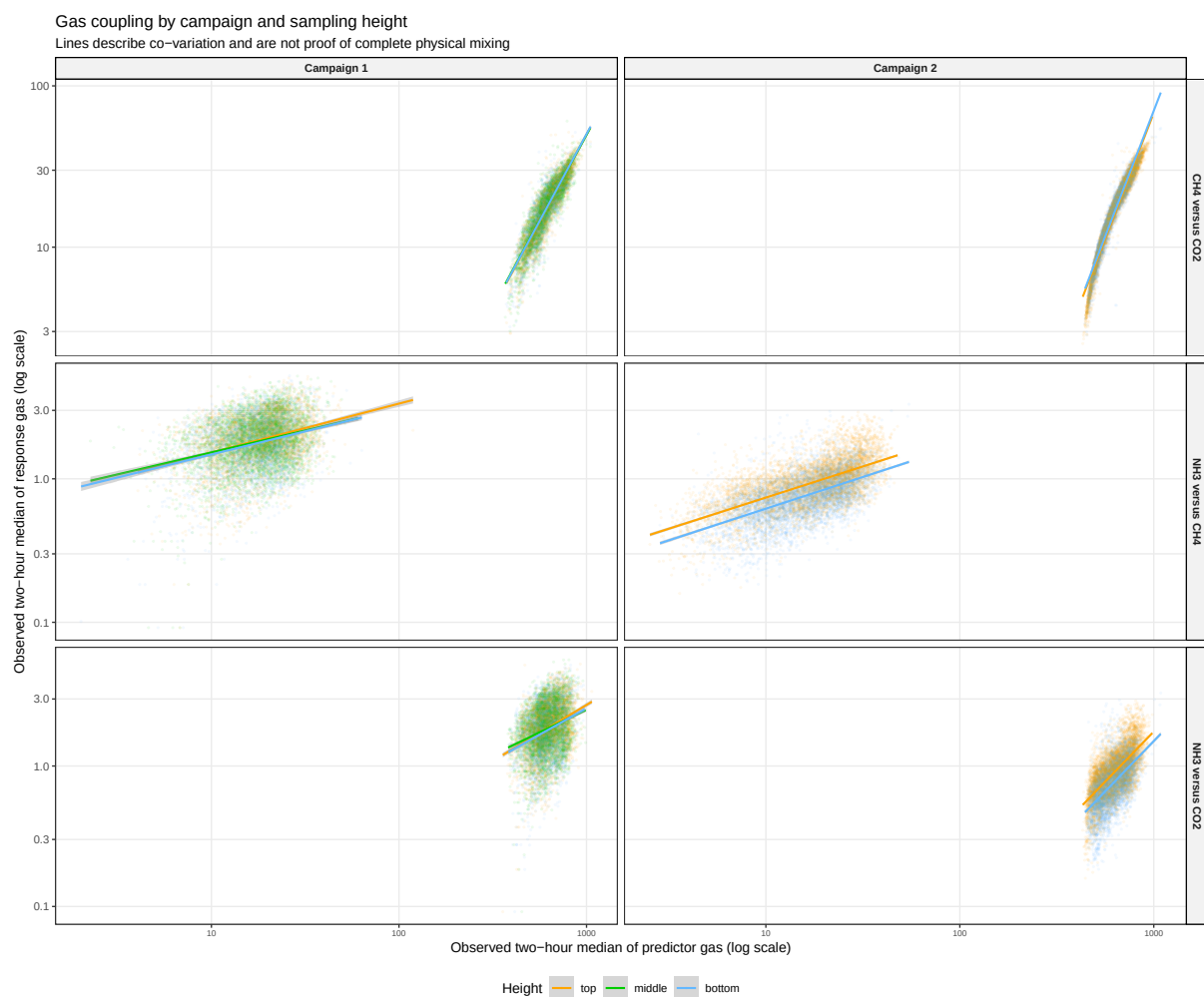
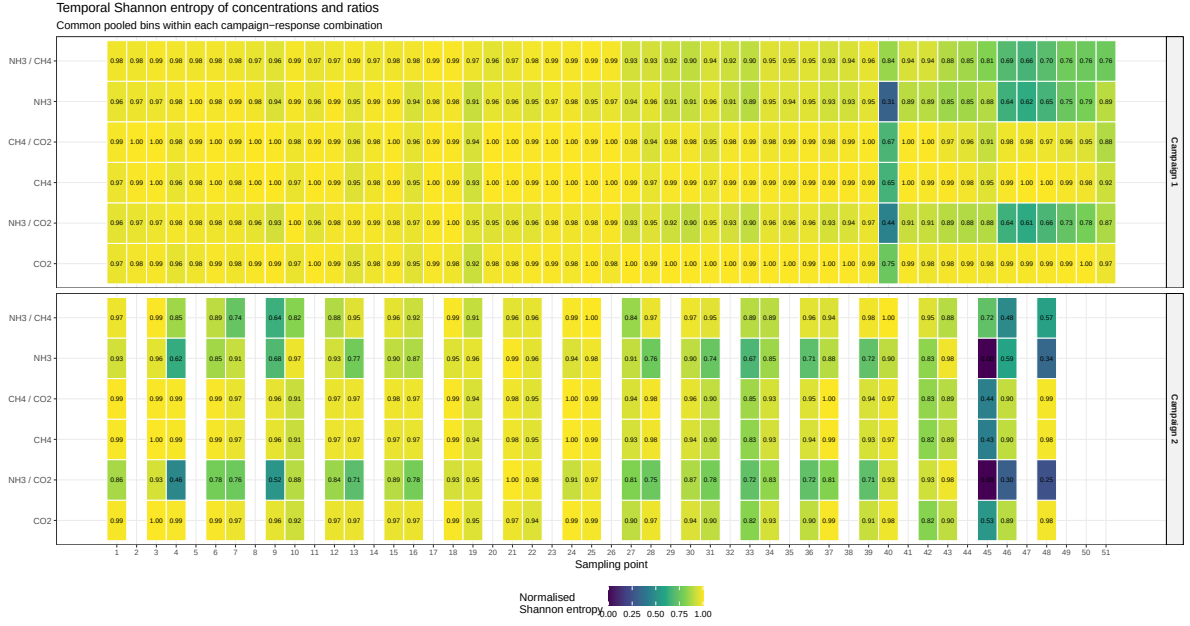
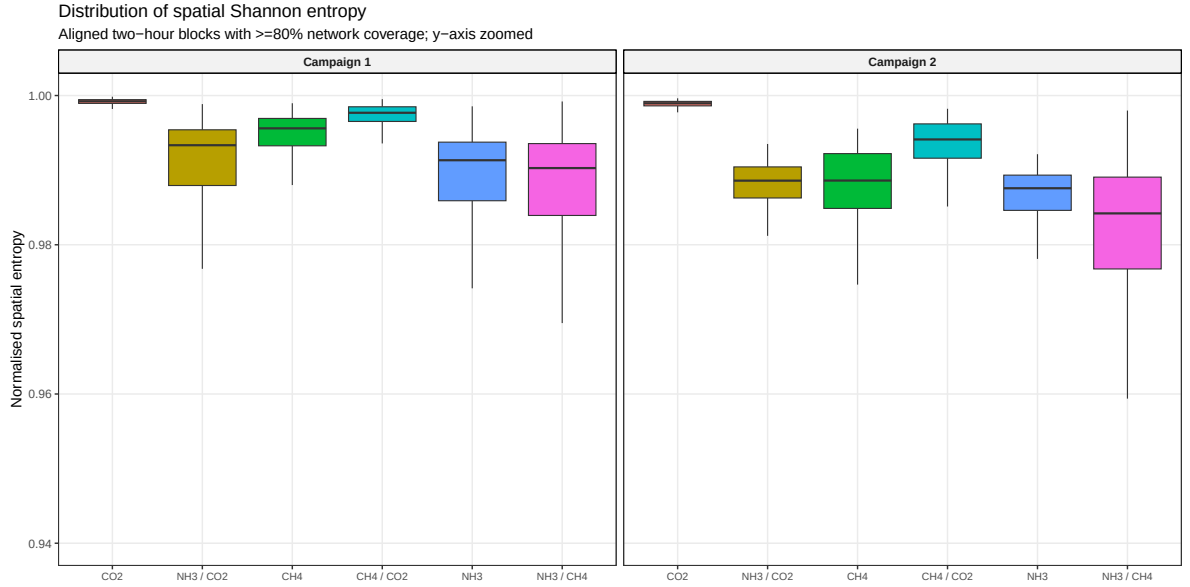


Figure 5: Observed gas coupling using two-hour SP medians on logarithmic axes. Lines are height-specific ordinary least-squares summaries. Covariation is a mixing diagnostic, not evidence that source distributions or transport histories are identical.



(a) Temporal entropy at each SP in both campaigns.



(b) Distribution of spatial entropy among contemporaneous SPs.

Figure 6: Temporal and spatial four-bin Shannon entropy for concentrations and ratios. Higher temporal entropy denotes more even occupancy of concentration states at a location; higher spatial entropy denotes more even occupancy across the sampled network at a time. Neither quantity is an estimate of airflow or measurement error.

Table 1: Descriptive statistics for daily whole-network estimates and sampling point (SP) 25. Gas ratios are dimensionless and are not multiplied by 100. Whole-network daily estimates weight each available SP equally. Relative error (RE) is signed for SP 25 against the contemporaneous network mean.

CO ₂					NH ₃ /CO ₂				
Statistic	C1 network	C1 SP25	C2 network	C2 SP25	Statistic	C1 network	C1 SP25	C2 network	C2 SP25
Mean	619.66	616.75	638.52	643.93	Mean	0.00386	0.00332	0.00139	0.00148
SD	47.11	56.01	84.67	91.76	SD	0.00580	0.00068	0.00024	0.00024
CV (%)	7.60	9.08	13.26	14.25	CV (%)	150.36	20.56	17.51	16.38
Median	615.25	610.67	640.47	652.74	Median	0.00331	0.00327	0.00134	0.00143
MAD	36.86	40.40	76.15	69.82	MAD	0.00050	0.00052	0.00013	0.00015
Minimum	501.60	470.99	489.69	484.90	Minimum	0.00158	0.00166	0.00108	0.00112
Q1	582.23	575.34	562.83	555.29	Q1	0.00273	0.00284	0.00123	0.00129
Q3	658.40	663.17	711.44	713.82	Q3	0.00375	0.00389	0.00150	0.00160
Maximum	740.94	767.14	797.24	834.89	Maximum	0.06441	0.00483	0.00212	0.00206
95% CI lower	610.92	606.36	613.38	616.68	95% CI lower	0.00278	0.00319	0.00132	0.00141
95% CI upper	628.41	627.15	663.67	671.18	95% CI upper	0.00494	0.00344	0.00147	0.00155
Daily estimates	114	114	46	46	Daily estimates	114	114	46	46
Measurement days	114	114	46	46	Measurement days	114	114	46	46
SP25 RE (%)	0.00	-0.56	0.00	0.74	SP25 RE (%)	0.00	0.98	0.00	6.86

CH ₄					CH ₄ /CO ₂				
Statistic	C1 network	C1 SP25	C2 network	C2 SP25	Statistic	C1 network	C1 SP25	C2 network	C2 SP25
Mean	18.70	19.52	18.53	19.01	Mean	0.02960	0.03059	0.02740	0.02790
SD	2.87	3.58	6.65	7.33	SD	0.00506	0.00322	0.00679	0.00749
CV (%)	15.33	18.34	35.86	38.56	CV (%)	17.10	10.53	24.78	26.86
Median	18.49	19.17	18.55	19.69	Median	0.02956	0.03054	0.02772	0.02897
MAD	2.08	2.58	5.80	6.07	MAD	0.00156	0.00209	0.00557	0.00610
Minimum	9.73	9.12	7.29	6.79	Minimum	0.01906	0.01916	0.01457	0.01376
Q1	16.64	16.78	12.26	12.45	Q1	0.02777	0.02852	0.02117	0.02117
Q3	20.83	21.98	24.07	24.35	Q3	0.03098	0.03264	0.03294	0.03375
Maximum	27.37	30.12	30.86	34.32	Maximum	0.07539	0.03916	0.03792	0.04021
95% CI lower	18.17	18.86	16.56	16.84	95% CI lower	0.02866	0.02999	0.02539	0.02567
95% CI upper	19.23	20.19	20.51	21.19	95% CI upper	0.03054	0.03119	0.02942	0.03012
Daily estimates	114	114	46	46	Daily estimates	114	114	46	46
Measurement days	114	114	46	46	Measurement days	114	114	46	46
SP25 RE (%)	0.00	4.09	0.00	1.55	SP25 RE (%)	0.00	4.13	0.00	1.25

NH ₃					NH ₃ /CH ₄				
Statistic	C1 network	C1 SP25	C2 network	C2 SP25	Statistic	C1 network	C1 SP25	C2 network	C2 SP25
Mean	1.99	2.00	0.90	0.96	Mean	0.12497	0.11828	0.05816	0.06182
SD	0.40	0.39	0.25	0.24	SD	0.03718	0.02986	0.01401	0.02006
CV (%)	19.98	19.60	28.08	24.83	CV (%)	29.75	25.24	24.10	32.45
Median	2.01	1.99	0.84	0.90	Median	0.12621	0.11830	0.05723	0.05714
MAD	0.27	0.26	0.16	0.15	MAD	0.02288	0.02323	0.00968	0.01476
Minimum	0.89	0.92	0.53	0.60	Minimum	0.07046	0.05895	0.03216	0.03065
Q1	1.70	1.72	0.73	0.79	Q1	0.09719	0.09451	0.04761	0.04473
Q3	2.23	2.25	1.04	1.09	Q3	0.14548	0.13849	0.06684	0.07464
Maximum	3.23	3.08	1.63	1.58	Maximum	0.30039	0.18517	0.08961	0.11027
95% CI lower	1.92	1.93	0.83	0.89	95% CI lower	0.11807	0.11274	0.05400	0.05587
95% CI upper	2.07	2.08	0.98	1.03	95% CI upper	0.13187	0.12382	0.06232	0.06778
Daily estimates	114	114	46	46	Daily estimates	114	114	46	46
Measurement days	114	114	46	46	Measurement days	114	114	46	46
SP25 RE (%)	0.00	1.02	0.00	7.28	SP25 RE (%)	0.00	-3.39	0.00	4.61

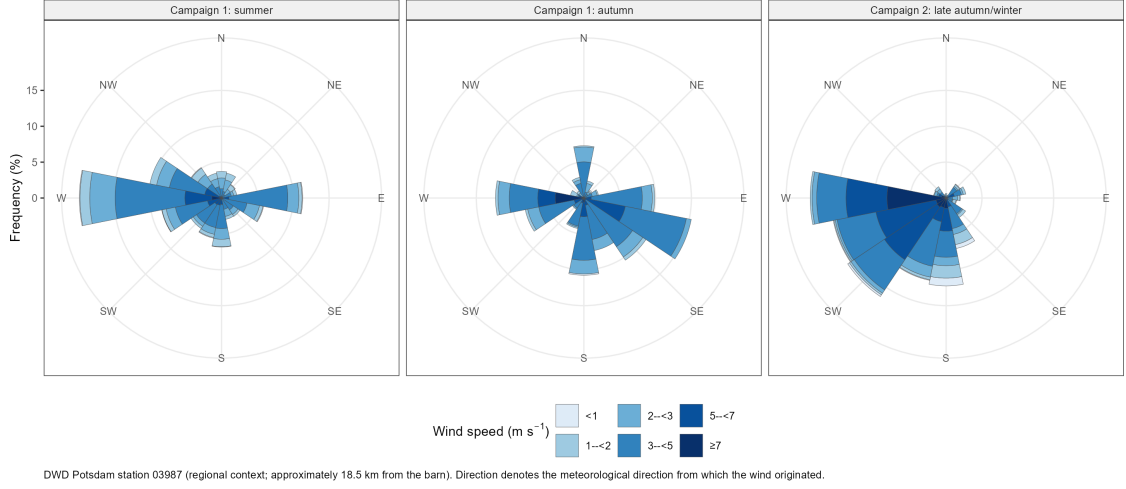


Figure 7: Seasonal wind direction and speed from the German Meteorological Service. These regional measurements describe seasonal forcing and were not treated as direct measurements of airflow through the barn.

Table 2: Omnibus tests from the linear mixed-effects models. The dense model used Campaign 1 and all three heights; the shared model used top and bottom SPs available across the four seasonal periods. Models were fitted to log-transformed responses with horizontal position and date as random intercepts.

Response	Campaign 1 dense			Shared top-bottom		
	Height	Period	Height $\times$ period	Height	Period	Height $\times$ period
CO ₂	< 0.001	< 0.001	< 0.001	< 0.001	0.002	< 0.001
NH ₃ /CO ₂	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
CH ₄	< 0.001	< 0.001	< 0.001	< 0.001	0.008	< 0.001
CH ₄ /CO ₂	< 0.001	0.053	< 0.001	< 0.001	< 0.001	< 0.001
NH ₃	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
NH ₃ /CH ₄	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001

3.5 Propagation of spatial sampling into ventilation and emissions

Of the matched Campaign 1 blocks, 473 satisfied the conservative primary requirement $\Delta\text{CO}_2 \geq 200$ ppm. For the full 51-SP network, median standard-input ventilation was $960 \text{ m}^3 \text{ h}^{-1} \text{ LU}^{-1}$. Relative to this reference, the 34-point top-and-bottom configuration produced median absolute errors of 1.16% for Q , 1.44% for CH_4 emission and 2.05% for NH_3 emission. The Campaign-2-like 32-point coverage produced 1.74, 1.55 and 2.83%, respectively. A single middle height was the best 17-point configuration (1.74, 2.41 and 3.31%), whereas top-only and bottom-only networks generated larger and more variable errors, particularly for NH_3 (Fig. 8).

Random balanced networks showed a monotonic density effect. Median network errors declined from approximately 8–9% for Q , 4–5% for CH_4 emission and 24% for NH_3 emission at three SPs to below 2% for Q and CH_4 , and approximately 5% for NH_3 , at 36 SPs. The much stronger density sensitivity of NH_3 is consistent with its local sources and weaker coupling with CO_2 .

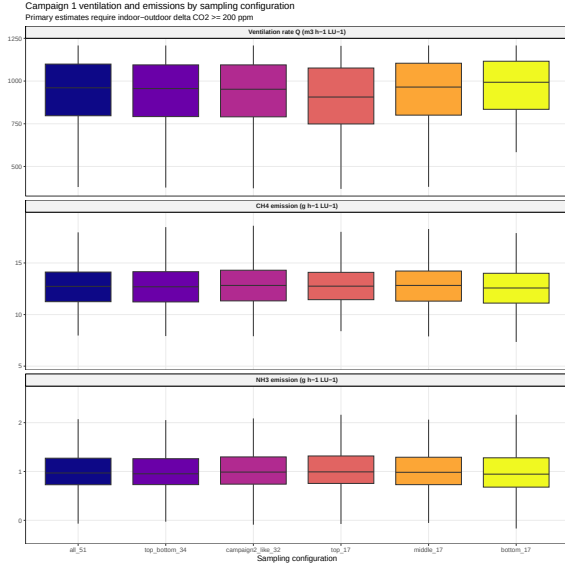
Gas ratios were associated with calculated rates, but shared numerator and denominator terms create algebraic coupling. Spearman ρ was 0.56 between CH_4/CO_2 and CH_4 emission, 0.79 between NH_3/CO_2 and NH_3 emission, and 0.77 between NH_3/CH_4 and NH_3 emission. These ratios are useful diagnostics of relative gas behaviour but are not independent predictors of emissions.

4 Discussion

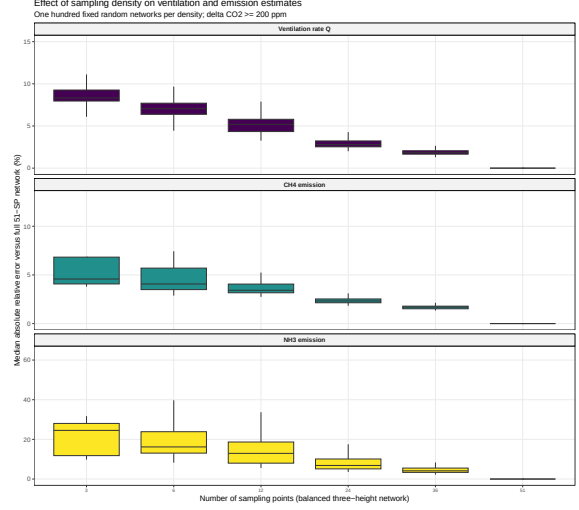
4.1 Precision is a property of gas, space and time

The first hypothesis was strongly supported. A statement such as “the concentration was spatially variable” is incomplete unless gas and aggregation interval are specified. CO_2 had the lowest hourly CV, consistent with a broadly distributed animal source and a relatively large ambient baseline. NH_3 and CH_4 retained greater spatial variability, consistent with local manure, animal and activity sources. Aggregation reduced much of the transient variability but did not eliminate gas-specific differences. This explains how campaign means may appear stable while hourly sensor placement remains consequential for emission calculations.

The result complements the direct emission analysis of Janke et al. (2020b): their large error at one constant height arose from jointly unresolved concentration and velocity profiles, whereas the present work isolates the concentration component in an occupied barn. The decrease in CV with aggregation should therefore not be interpreted as proof that a temporally averaged single concentration is flux-representative. Concentration and



(a) Absolute rates by height and coverage configuration.



(b) Error relative to the 51-SP reference for 100 balanced networks per density.

Figure 8: Propagation of spatial sampling into standard-input indirect CO_2 -balance ventilation and emission estimates. The primary analysis requires $\Delta\text{CO}_2 \geq 200$ ppm. These matched-block comparisons isolate spatial sampling effects under common calculation assumptions; they are not validation against direct airflow measurements.

Table 3: Campaign 1 sampling-configuration effects relative to the full 51-SP network. Values are median absolute relative error; parentheses contain the 95th percentile.

Configuration	SPs	Q (%)	CH_4 emission (%)	NH_3 emission (%)
Top only	17	6.50 (16.99)	2.74 (8.94)	6.17 (38.81)
Middle only	17	1.74 (7.61)	2.41 (7.33)	3.31 (18.90)
Bottom only	17	5.14 (18.23)	3.04 (9.95)	5.43 (36.06)
Top + bottom	34	1.16 (5.33)	1.44 (5.23)	2.05 (12.42)
Campaign-2-like	32	1.74 (7.59)	1.55 (6.31)	2.83 (16.06)

velocity must ultimately be sampled at matching spatial and temporal resolutions.

4.2 The vertical-increase hypothesis was conditional

The near-roof increase reported by Sahu et al. (2023) was reproduced most clearly in autumn, but only weakly in summer and not at every horizontal position. This partial support is scientifically more informative than a universal confirmation. It agrees with the gas-specific profiles of D’Urso et al. (2024) and with the simulation result of Doumbia et al. (2021) that an approximately 3 m line represents only some flow regimes. Roof geometry, buoyant animal plumes, fan operation, wind-driven short-circuiting and local sources can each alter the vertical ordering.

The middle SPs at 3.6 m were close to the common measurement height and only 1 m above the bottom SPs. The dense design was valuable because it bracketed that conventional region with a bottom layer at 2.6 m and an uncommon near-roof layer. Contrary to a redundancy interpretation, the middle height was the most representative single-height estimate for every concentration and ratio. It was removed only after the FTIR failures forced a 32-port CRDS design. Retaining both extremes nevertheless preserved the observed vertical contrast in Campaign 2 and reproduced the dense-network balance more closely than either extreme alone. This operational result should not be generalised to barns with different roof slopes, opening geometry, manure management or mechanical air mixing.

Ratio gradients reinforce this interpretation. If vertical differences reflected only uniform dilution, the three ratios would have been stable. Their significant height-by-period interactions instead indicate differing source distributions, chemical behaviour or transport histories. NH_3 is particularly sensitive to temperature, manure surface conditions and deposition, whereas CH_4 and CO_2 arise from overlapping but non-identical animal and manure processes. The ratios are therefore diagnostic, not substitutes for tracer-based ventilation or mass-balance analysis.

This interpretation partly agrees with, but also extends, the spatial-ratio experiment of Mendes et al. (2015). In their naturally ventilated dairy barn, the vertical variability of NH_3 , metabolically produced CO_2 , injected SF_6 , and their ratios decreased with height and became approximately constant above the animal-occupied zone (greater than approximately 2 m). They consequently recommended sampling NH_3/CO_2 above that zone. All three heights in the present study were above 2 m, yet NH_3/CO_2 still showed height- and season-dependent structure, and the near-roof layer was not interchangeable with the 2.6 m or 3.6 m layers. The difference is plausible because Mendes et al. used short, approximately two-hour trials under comparatively stable winds, two transverse sampling poles and controlled curtain openings, whereas the present data covered 17 horizontal positions, uncommon near-roof locations and changing summer-to-winter forcing. Mendes

et al. also cautioned that longitudinal variability, ridge sampling, changing wind and window opening required further study; those unresolved dimensions are directly represented, although not experimentally controlled, here.

The gas-coupling results clarify why the two studies should not be read as contradictory. The strong CH_4 – CO_2 coupling indicates broadly shared dilution and overlapping animal sources, while the weak Campaign 1 NH_3 – CO_2 coupling and spatially structured NH_3/CO_2 ratio indicate that metabolic CO_2 did not reproduce ammonia behaviour equally at every SP. Mendes et al. similarly found that homogeneous pollutant and tracer concentrations were not required for stable ratios, but only when their dispersion patterns remained sufficiently similar. Taken together, the studies support ratio sampling above the animal zone as a useful starting principle, not a guarantee of barn-wide representativeness across roof geometry, season and pollutant.

4.3 Wind, neighbouring sources and sensor placement

Westerly periods did not produce a general concentration rise in these data. Their lower internal means suggest that ventilation enhancement dominated during the observed hours, although this inference remains provisional. A neighbouring naturally ventilated dairy barn could contribute intermittent plumes when upwind, but regional meteorology and internal concentration alone cannot identify that source. A future study should measure approach flow and gas concentrations between the two barns and explicitly test how inter-barn distance, atmospheric stability and plume meandering affect inlet concentrations.

Entropy was response- and height-specific. In Chen et al. (2026), it quantified the sensitivity of simulated airflow variables to prescribed wind cases; here, it quantified temporal occupancy of measured concentration states shaped by sources and ventilation. The values are therefore conceptually related but not directly transferable, and high entropy alone does not identify a universally representative SP.

4.4 Consequences for ventilation and emission estimation

The spatial sensitivity experiment demonstrates why concentration heterogeneity matters beyond descriptive mapping. A reduced network containing both vertical extremes reproduced the 51-SP Q and emission estimates with small median error, whereas top-only and bottom-only networks introduced systematic coverage error. Balanced density reduction affected NH_3 emission much more strongly than CH_4 , consistent with the weak NH_3 – CO_2 coupling and localised ammonia sources. These findings complement Janke et al. (2020b): their direct measurements showed that unresolved concentration–velocity profiles can cause large flux errors, whereas the present matched-block exercise isolates the concentration-sampling component.

The ΔCO_2 rule is important. Since Q is inversely proportional to indoor–outdoor

CO₂ difference, weak enhancements amplify analytical and spatial errors. The 200 ppm primary threshold was therefore used for conservative inference, with 70–199 ppm retained for sensitivity analysis and values below 70 ppm flagged. This does not make the calculated absolute rates a validation dataset. Animal heat and CO₂ production were represented by standard inputs, short-period animal activity was not independently measured, and airflow at the openings was unavailable. The robust conclusion is the relative effect of applying alternative spatial concentration networks to the same blocks and assumptions.

4.5 Limitations and future work

Sequential sampling trades spatial coverage against simultaneity. Although two-hour complete blocks and mixed models reduced this confounding, rapidly changing flow can evolve within a switching cycle. Campaign and season were also partly aliased with network reduction and analyser deployment. DWD measurements were not co-located at the openings. Finally, retaining every positive concentration avoids an arbitrary physical threshold but makes ratio moments sensitive to small denominators; robust block-level summaries are therefore essential.

Future measurements should deploy three-component velocity sensors at spatial and temporal resolutions matched to the gas measurements, especially at vertical openings and near the roof. The measured concentration fields can then constrain CFD simulations or wind-tunnel experiments that resolve inflow, outflow and recirculation. Such work would permit emission-weighted optimisation rather than concentration-only optimisation, and would test whether the reduced network remains valid across barns, seasons and neighbouring-source configurations.

5 Conclusions

The apparent precision of barn gas concentration measurements depended jointly on gas, spatial coverage and temporal aggregation. Hourly spatial CV was consistently lowest for CO₂ and substantially greater for CH₄ and NH₃; averaging reduced but did not remove this distinction. Concentrations generally increased from bottom to top in autumn and winter, but the ordering was weaker and spatially inconsistent in summer. Accordingly, H1 was supported, whereas H2 was only conditionally supported.

The middle layer at 3.6 m was the most representative single height, while no single SP represented every gas and ratio; SP17 ranked first and the pragmatic central SP25 ranked fourth. The middle layer was omitted from Campaign 2 because FTIR failure required a 32-port CRDS network, not because it was statistically dispensable. Ratios, observed gas-coupling plots and temporal and spatial entropy maps all demonstrated gas-specific behaviour.

Using the 51-SP Campaign 1 network as reference, retaining both top and bottom heights limited median errors in standard-input ventilation and emission calculations to approximately 1–3%, whereas a single extreme height and very sparse networks were less reliable, especially for NH_3 . These results quantify the concentration-sampling component only. Future studies should pair gas and three-component velocity measurements at matching spatio-temporal resolution and use the resulting fields to validate CFD or wind-tunnel optimisation.

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Data and code availability

The reproducible R workflow used for the reported analysis is version controlled with the manuscript. Raw instrument data are retained by the authors and can be made available subject to institutional data-management requirements.

A Supplementary meteorological context

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Daily summaries from DWD Potsdam station 03987. Shaded temperature bands show the daily minimum–maximum range. Missing intervals between study periods are intentionally retained.

Figure 9: Daily air temperature, relative humidity and precipitation from the DWD Potsdam station. Monthly ticks run from June to December 2024. Missing intervals between campaigns are retained.

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